# Zhiqian Chen

zhiqian.chen@rit.edu
https://zhiqian.ai
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RESEARCH Fundamental principles of networked system dynamics by integrating machine learning, network science, and dynamical systems.

## Employment

2026-now **Rochester Institute of Technology**, Assistant Professor
Computer Science, Golisano College of Computing and Information Sciences

2020-2026 **Mississippi State University**, Assistant Professor
Computer Science and Engineering, Bagley College of Engineering

## Education

2014–2020 **Virginia Tech**, *Ph.D., Computer Science*, Falls Church, Virginia, United States

2010–2013 **Peking University**, *M.E., Software Engineering*, China

2005–2009 **Huazhong Univ. of Sci. & Tech.**, *B.E., Software Engineering; B.A., Japanese (minor)*, China

## Grants

**Personal Credits: $ 1,734,101.** / **Total External: $ 11,436,553.**

**NSF** CAREER: From Fragmentation to Integration: Advancing Cross-Graph Dynamics in Interdependent Networks (#2443266) NSF award link

Sole PI, **$517,594** (100% credit), 06/2025-05/2030

**NSF** CISE MSI: RCBP: III: Advancing Speech Detection: A Hybrid Approach Using Large Language Models and Graph Neural Networks (#2431176) NSF award link

Co-PI/University PI, **$400,000** (33% credit, MS State share $259,948.00), 1/2025-12/2026

**NSF** CIRC: Planning-C: Synergistic Graph Flow Analytics: An Integrated Infrastructure for Bridging Complexity, Fragmentation, and Interdisciplinary Gaps (#2345921) NSF award link

PI, **$99,998** (60% credit), 7/2024-6/2025

**NSF** CRII: Interpretable Influence Propagating and Blocking on Graphs (#2153369) NSF award link

Sole PI, **$174,004** (100% credit), 5/2022-11/2024

**USDA** [ARS] Developing Detection and Modeling Tools for the Geospatial and Environmental Epidemiology of Animal Disease (#58-6064-3-017) Project Page

Co-PI, **$3,073,602** (5% credit), 10/2023-08/2028

**USDA** [ARS] Advancing Agricultural Research through High Performance Computing (#58-0200-0-002)

Co-PI, **$5,690,689** (5% credit), 10/2022-08/2024

**NSF** [EDU] ITEST: Learning to create Intelligent Solutions with Machine Learning and Computer Vision: A Pathway to AI Careers for Diverse High School Students (#2342574) NSF award link

Co-PI, **$1,408,467** (25% credit), 09/2024-08/2027

**Turing AI** Gift Money, **$25,000** (100% credit), 08/2025

**NSF** [III] REU Supplementary, PI, **$14,000** (100% credit), 11/2023-11/2024

**USDA** [ARS] Summer Internship Project, PI, **$8,199** (100% credit), 5/2024-4/2025

**Google** Google Cloud Research Credits, sole-PI, **$5,000** (100% credit), 11/2024-11/2026

**CRA** REU, sole-PI, **$10,000.0** (100% credit), 5/2026-8/2 026

**NAIRR** Start-up project, sole-PI, **2,000.0 GPU Hours** , 2/2026-5/2026 026

**Total Internal $ 19,094.5**

CTL PI, Ottilie Schillig Special Teaching Award, $2,994.50 Awardee List

INTER-INS PI, Global Development Seed Grant Award, International Institute $6,000 News Page

BAGLEY PI, Working Group in Graph AI, Bagley College of Engineering $4,100

ORED PI/Co-PI, Undergrad Research Program, 2022/2024/2025 (Sole PI, $2,000), 2023 (Co-PI, $1,500)

## Awards & Honors

AWARD Best Paper Award at ACM SIGSPATIAL 2020 Awards Recipients

AWARD Best Paper Award at GISTAM 2015 Awards Recipients

AWARD Outstanding Contribution Award, 2016, Toyota Research Institute, North America

AWARD Ottilie Schillig Special Teaching Award, 2025, Center for Teaching and Learning Award Recipients

HONOR Editor's Choice Article Editor Choices

HONOR Top 50 Chem. & Mater. Sciences Articles in **Nature Communications** 2019 Top Paper List

HONOR Top Area Chair, NeurIPS, 2026

HONOR Excellent Reviewer, IEEE Transactions on Network Science and Engineering, 2024

HONOR Excellent Reviewer (top 20%), SIG KDD, 2025

## Talks

Tutorial Uncertainty Quantification for Dynamical Networks
**WSDM 2025**, Feb 22-26, Boise Idaho U.S. Tutorial page at SDM

Tutorial Unifying Spectral and Spatial Graph Neural Network
**SDM 2025**, May 3, D.C., U.S. Tutorial page at SDM

Tutorial Unifying Spectral and Spatial Graph Neural Network
**CIKM 2024**, Oct 21, Boise, Idaho, U.S. CIKM tutoiral list

Tutorial Unifying Spectral and Spatial Graph Neural Network
**SIAM Math & Data Science 2024**, Oct 23, Atlanta, GA, U.S. Tutorial page at MDS

Tutorial Unifying Spectral and Spatial Graph Neural Network
**CVPR 2024**, June 18, Seattle, Washington, U.S. Tutorial page at CVPR

Paper Early forecasting of the impact of traffic accidents using a single shot observation
**SIAM Data Mining 2022**, April, Minneapolis, Minnesota, U.S.

Tutorial Studying spread patterns of covid-19 based on spatiotemporal data
**SIAM Data Mining 2021**, April, Online

## Teaching

Undergrad CSE 4633/6633 Artificial Intelligence, CSE 4693/6693 Intro to Machine Learning, CSCI 432 Intro to Computer Vision.

Graduate CSE 8673 Machine Learning, CSE 8990 Graph Machine Learning

## Mentoring / as major research advisor

Ph.D. Marouane Benbrahim, Tamanna Rashme, Shaowen Wang

## Alumni

PhD Zonghan Zhang Graduated at Fall 2025, Amin George Graduated at Spring 2026, Josh Waldbieser, Xinyuan Chen, Zijian Zhang, Jiashan Wu, Rocker D'Antonio.

Master Matthew Rester, Henry Smith, Samuel Prabhakar, Rajeev Jogi, Piero Bracamonte, Suman Adhikari, Ramyasri Veerapaneni, Aymane Jouhari, Sai Harika Gade, Pavan Dharma Adapa .

Undergrad Kevin Ho, Jason Weeks, Ben Moore, Jack Maloney, Ethan Rogers, Prathyusha Mustiyala, Andrew McBride, Aalok Uprety, Mason Fisher, Reid Sewell.

## Service

Co-Chair IEEE ICDM 2025, Poster Organization Committee

Co-Chair IEEE BigData 2026, Student Volunteer Organization Committee

Co-Chair IEEE BigData 2024, BigData Cup Challenges Organization Committee

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|---|---|
| Panelist | NSF Panel, 2021, 2023, 2024 |
| Editorial Board | Frontiers in Big Data - Data Science, Frontiers in Big Data - Data Mining and Management, Frontier Topic Editor, 2022-present |
| Area Chair | Neural Information Processing Systems (*NeurIPS*), 2026 |
| Reviewer | International Conference on Machine Learning (*ICML*). International Conference on Learning Representations (*ICLR*). Neural Information Processing Systems (*NeurIPS*). AAAI Conference on Artificial Intelligence (*AAAI*). International Joint Conference on Artificial Intelligence (*IJCAI*). ACM SIG on Knowledge Discovery and Data Mining (*KDD*). ACM SIG on Information Retrieval (*SIGIR*). IEEE Transactions on Knowledge and Data Engineering (*TKDE*). ACM Transactions on Knowledge Discovery from Data (*TKDD*) |
| Advisor | Mississippi State University AI club Student Organizations in the Bagley College of Engineering |

## Publications


[1] Mariam Almutairi, Fatimah Alotaibi, Lulwah Alkulaib, Zhiqian Chen, Sara Alsalamah, and Chang-Tien Lu. Query-graph compatibility: A formal pre-condition for faithful knowledge-grounded language model generation. In *Proceedings of the 35th ACM International Conference on Information and Knowledge Management*, 2026. Accepted, to appear.

[2] Marouane Benbrahim, Xin Fang, and Zhiqian Chen. Sumo x pypsa: Interactive web demo of real-time urban power-traffic co-simulation with vehicle-to-grid. In *WWW Companion '26: Companion Proceedings of the ACM Web Conference 2026*, pages 180–183. Association for Computing Machinery, 2026.

[3] Marouane Benbrahim, Kavya Gautam, Zonghan Zhang, and Zhiqian Chen. Llm-to-map: Transparent conversational tool orchestration for real-time multi-domain simulation. In *Proceedings of the Thirty-Fifth International Joint Conference on Artificial Intelligence (IJCAI-26)*, 2026. Accepted, to appear.

[4] Marouane Benbrahim, Rayane Goghrod, Tamanna Rashme, Taoran Ji, Kaiqun Fu, Lei Zhang, and Zhiqian Chen. Delta-graph signatures for post-edit risk screening in knowledge editing. In *Proceedings of the 2026 SIAM International Conference on Data Mining (SDM)*. Society for Industrial and Applied Mathematics, 2026. Accepted, to appear.

[5] Zhiqian Chen and Zonghan Zhang. Uncertainty quantification for dynamical networks. In *Proceedings of the Nineteenth ACM International Conference on Web Search and Data Mining*, pages 1355–1358, 2026.

[6] Md Toyaha Rahman Ratul, Zhiqian Chen, Kaiqun Fu, Taoran Ji, and Lei Zhang. Masterset: A large-scale benchmark for must-cite citation recommendation in the ai/ml literature. In *Proceedings of the 2026 SIAM International Conference on Data Mining (SDM)*. Society for Industrial and Applied Mathematics, 2026. Accepted, to appear.

[7] Tiantian Yang and Zhiqian Chen. Tf-dwgnet: A directed weighted graph neural network with tensor fusion for multi-omics cancer subtype classification. *NAR Genomics and Bioinformatics*, 8(2):lqag054, 2026.

[8] Andrew Meng and Zhiqian Chen. Deep gaussian fusion network for traffic prediction. In *2025 IEEE International Conference on Data Mining Workshops (ICDMW)*, pages 3046–3050. IEEE, 2025.

[9] Tamanna Rashme, Zonghan Zhang, Jason Weeks, Marouane Benbrahim, Zijian Zhang, Zhiqian Chen, Nisha Pillai, Ram Ramkumar, and Bindu Nanduri. Graph symbolic regression to interpret the propagation of vesicular stomatitis virus across the us and mexico. In *Proceedings of the 33rd ACM International Conference on Advances in Geographic Information Systems*, pages 977–980, 2025.

[10] Adil Saleem, Leon L Shaw, Zhiqian Chen, and Wei Lai. State-of-the-art machine learning technology for sustainable lithium battery cathode design: A perspective. *Advanced Energy Materials*, 15(31):2405300, 2025.

[11] Lei Zhang, Zhiqian Chen, Chang-Tien Lu, and Liang Zhao. Network interdiction goes neural. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2*, pages 3774–3785, 2025.

[12] Lei Zhang, Qisheng Zhang, Zhiqian Chen, Yanshen Sun, Chang-Tien Lu, and Liang Zhao. Implicit graph neural network for deep graph transformation. *Knowledge and Information Systems*, pages 1–30, 2025.

[13] Zonghan Zhang and Zhiqian Chen. Sobol's total indices for accurate and scalable feature exclusion in high-dimensional data. In *2025 IEEE International Conference on Big Data (BigData)*, pages 1317–1326. IEEE, 2025.

[14] Mariam ALMutairi, Lulwah AlKulaib, Shengkun Wang, Zhiqian Chen, Youssif ALMutairi, Thamer M Alenazi, Kurt Luther, and Chang-Tien Lu. Fhirviz: Multi-agent platform for fhir visualization to advance healthcare analytics. In *Proceedings of the 15th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics*, pages 1–7, 2024.

[15] Zhiqian Chen, Lei Zhang, and Liang Zhao. Unifying spectral and spatial graph neural networks. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*, pages 5511–5513, 2024.

[16] Taoran Ji, Nathan Self, Kaiqun Fu, Zhiqian Chen, Naren Ramakrishnan, and Chang-Tien Lu. Citation forecasting with multi-context attention-aided dependency modeling. *ACM Transactions on Knowledge Discovery from Data*, 18(6):1–23, 2024.

[17] Kollin Napier, Tanmay Bhowmik, and Zhiqian Chen. Explaining poor performance of text-based machine learning models for vulnerability detection. *Empirical Software Engineering*, 29(5):113, 2024.

[18] Nisha Pillai, Ganga Gireesan, Michael James Rothrock Jr, Bindu Nanduri, Zhiqian Chen, and Mahalingam Ramkumar. Explainable chickenomics: Advancing poultry ai with interpretability. In *Plant and Animal Genome Conference (PAG 31)*. PAG, 2024.

[19] Nisha Pillai, Bindu Nanduri, Michael J Rothrock Jr, Zhiqian Chen, and Mahalingam Ramkumar. Bayesian-guided generation of synthetic microbiomes with minimized pathogenicity. In *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pages 1–7. IEEE, 2024.

[20] Rui Shang, Siji Chen, Zhiqian Chen, and Chang-Tien Lu. Graphnilm: A graph neural network for energy disaggregation. In *Pacific-Asia Conference on Knowledge Discovery and Data Mining*, pages 431–443. Springer Nature Singapore Singapore, 2024.

[21] Jiashan Wu and Zhiqian Chen. Dual-approach interpretations of bailout strategies in the eisenberg-noe model. In *World Congress in Computer Science, Computer Engineering & Applied Computing*, pages 398–411. Springer Nature Switzerland Cham, 2024.

[22] Zirui Yuan, Minglai Shao, and Zhiqian Chen. Graph bayesian optimization for multiplex influence maximization. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 22475–22483, 2024.

[23] Zijian Zhang, Zonghan Zhang, and Zhiqian Chen. Flowgpt: How long can llms trace back and predict the trends of graph dynamics? In *SouthNLP*, 2024.

[24] Zijian Zhang, Zonghan Zhang, and Zhiqian Chen. Neural tangent bayesian optimization for accurate and efficient influence maximization. In *2024 IEEE 36th International Conference on Tools with Artificial Intelligence (ICTAI)*, pages 806–813. IEEE, 2024.

[25] Zonghan Zhang, Ramyasri Veerapaneni, Moses Ayoola, Athish Ram Das, Zhiqian Chen, Bindu Nanduri, and Mahalingam Ramkumar. Leveraging graph neural networks for mic prediction in antimicrobial resistance studies. In *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pages 1–4. IEEE, 2024.

[26] Zonghan Zhang, Zijian Zhang, and Zhiqian Chen. Multiple-source localization from a single-snapshot observation using graph bayesian optimization. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 22538–22546, 2024.

[27] Yaya Zhao, Kaiqi Zhao, Zhiqian Chen, Yuanyuan Zhang, Yalei Du, and Xiaoling Lu. A graph-based representation framework for trajectory recovery via spatiotemporal interval-informed seq2seq. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI-24)*, pages 2588–2597. https://doi.org/10.24963/ijcai.2024/286, 2024.

[28] Kourosh T Baghaei, Amirreza Payandeh, Pooya Fayyazsanavi, Zhiqian Chen, Somayeh Bakhtiari Ramezani, and Shahram Rahimi. Deep representation learning: Fundamentals, technologies, applications, and open challenges. *IEEE Access*, 2023.

[29] Subhodip Biswas, Fanglan Chen, Zhiqian Chen, Chang-Tien Lu, and Naren Ramakrishnan. Memetic algorithms for spatial partitioning problems. *ACM Transactions on Spatial Algorithms and Systems*, 9(1):1–31, 2023.

[30] Fanglan Chen, Subhodip Biswas, Zhiqian Chen, Shuo Lei, Naren Ramakrishnan, and Chang-Tien Lu. Exploring tradeoffs in automated school redistricting: Computational and ethical perspectives. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 15912–15920, 2023.

[31] Zhiqian Chen, Fanglan Chen, Lei Zhang, Taoran Ji, Kaiqun Fu, Liang Zhao, Feng Chen, Lingfei Wu, Charu Aggarwal, and Chang-Tien Lu. Bridging the gap between spatial and spectral domains: A unified framework for graph neural networks. *ACM Computing Surveys*, 56(5):42, 2023.

[32] Nisha Pillai, Bindu Nanduri, Michael J Rothrock, Zhiqian Chen, and Mahalingam Ramkumar. Towards optimal microbiome to inhibit multidrug resistance. In *2023 IEEE Conference on Computational Intelligence in Bioinformatics and Computational Biology (CIBCB)*, pages 1–9. IEEE, 2023.

[33] Lei Zhang, Zhiqian Chen, Chang-Tien Lu, and Liang Zhao. Fast and adaptive dynamics-on-graphs to dynamics-of-graphs translation. *Frontiers in big Data*, 6:1274135, 2023.

[34] Lei Zhang, Qisheng Zhang, Zhiqian Chen, Yanshen Sun, Chang-Tien Lu, and Liang Zhao. Infinitely deep graph transformation networks. In *2023 IEEE International Conference on Data Mining (ICDM)*, pages 778–787. IEEE Computer Society, 2023.

[35] Zijian Zhang, Zonghan Zhang, and Zhiqian Chen. Xflow: Benchmarking flow behaviors over graphs. *arXiv preprint arXiv:2308.03819*, 2023.

[36] Zonghan Zhang and Zhiqian Chen. Accelerating simulation-based influence maximization via bayesian optimization. In *openreview.net*, 2023.

[37] Zonghan Zhang and Zhiqian Chen. Understanding influence maximization via higher-order decomposition. In *Proceedings of the 2023 SIAM International Conference on Data Mining (SDM)*, pages 766–774. Society for Industrial and Applied Mathematics, 2023.

[38] Subhodip Biswas, Fanglan Chen, Zhiqian Chen, Chang-Tien Lu, and Naren Ramakrishnan. Sampling-based techniques for designing school boundaries. *arXiv preprint arXiv:2206.03703*, 2022.

[39] Zhiqian Chen and Zonghan Zhang. Demystifying graph convolution with a simple concatenation. *arXiv preprint arXiv:2207.12931*, 2022.

[40] Guangyu Meng, Qisheng Jiang, Kaiqun Fu, Beiyu Lin, Chang-Tien Lu, and Zhiqian Chen. Early forecasting of the impact of traffic accidents using a single shot observation. In *Proceedings of the 2022 SIAM International Conference on Data Mining (SDM)*, pages 100–108. Society for Industrial and Applied Mathematics, 2022.

[41] Setareh Rafatirad, Houman Homayoun, Zhiqian Chen, and Sai Manoj Pudukotai Dinakarrao. Machine learning for computer scientists and data analysts: from an applied perspective, 2022.

[42] Setareh Rafatirad, Houman Homayoun, Zhiqian Chen, and Sai Manoj Pudukotai Dinakarrao. Adversarial machine learning. In *Machine Learning for Computer Scientists and Data Analysts: From an Applied Perspective*, pages 305–328. Springer International Publishing Cham, 2022.

[43] Setareh Rafatirad, Houman Homayoun, Zhiqian Chen, and Sai Manoj Pudukotai Dinakarrao. Applied machine learning for cloud resource management. In *Machine Learning for Computer*

*Scientists and Data Analysts: From an Applied Perspective*, pages 405–427. Springer International Publishing Cham, 2022.

[44] Setareh Rafatirad, Houman Homayoun, Zhiqian Chen, and Sai Manoj Pudukotai Dinakarrao. Applied machine learning for computer architecture security. In *Machine Learning for Computer Scientists and Data Analysts: From an Applied Perspective*, pages 383–404. Springer International Publishing Cham, 2022.

[45] Setareh Rafatirad, Houman Homayoun, Zhiqian Chen, and Sai Manoj Pudukotai Dinakarrao. A brief review of probability theory and linear algebra. *Machine Learning for Computer Scientists and Data Analysts: From an Applied Perspective*, pages 35–79, 2022.

[46] Setareh Rafatirad, Houman Homayoun, Zhiqian Chen, and Sai Manoj Pudukotai Dinakarrao. Graph learning. In *Machine Learning for Computer Scientists and Data Analysts: From an Applied Perspective*, pages 277–304. Springer International Publishing Cham, 2022.

[47] Setareh Rafatirad, Houman Homayoun, Zhiqian Chen, and Sai Manoj Pudukotai Dinakarrao. Online learning. In *Machine Learning for Computer Scientists and Data Analysts: From an Applied Perspective*, pages 235–256. Springer International Publishing Cham, 2022.

[48] Setareh Rafatirad, Houman Homayoun, Zhiqian Chen, and Sai Manoj Pudukotai Dinakarrao. Recommender learning. In *Machine Learning for Computer Scientists and Data Analysts: From an Applied Perspective*, pages 257–276. Springer International Publishing Cham, 2022.

[49] Setareh Rafatirad, Houman Homayoun, Zhiqian Chen, and Sai Manoj Pudukotai Dinakarrao. Reinforcement learning. In *Machine Learning for Computer Scientists and Data Analysts: From an Applied Perspective*, pages 217–232. Springer International Publishing Cham, 2022.

[50] Setareh Rafatirad, Houman Homayoun, Zhiqian Chen, and Sai Manoj Pudukotai Dinakarrao. Sensornet: An educational neural network framework for low-power multimodal data classification. In *Machine Learning for Computer Scientists and Data Analysts: From an Applied Perspective*, pages 331–357. Springer International Publishing Cham, 2022.

[51] Setareh Rafatirad, Houman Homayoun, Zhiqian Chen, and Sai Manoj Pudukotai Dinakarrao. Supervised learning. In *Machine Learning for Computer Scientists and Data Analysts: From an Applied Perspective*, pages 81–162. Springer International Publishing Cham, 2022.

[52] Setareh Rafatirad, Houman Homayoun, Zhiqian Chen, and Sai Manoj Pudukotai Dinakarrao. Transfer learning in mobile health. In *Machine Learning for Computer Scientists and Data Analysts: From an Applied Perspective*, pages 359–382. Springer International Publishing Cham, 2022.

[53] Setareh Rafatirad, Houman Homayoun, Zhiqian Chen, and Sai Manoj Pudukotai Dinakarrao. Unsupervised learning. In *Machine Learning for Computer Scientists and Data Analysts: From an Applied Perspective*, pages 163–216. Springer International Publishing Cham, 2022.

[54] Setareh Rafatirad, Houman Homayoun, Zhiqian Chen, and Sai Manoj Pudukotai Dinakarrao. What is applied machine learning? In *Machine Learning for Computer Scientists and Data Analysts: From an Applied Perspective*, pages 3–33. Springer International Publishing Cham, 2022.

[55] Jason Wang, Kaiqun Fu, Zhiqian Chen, and Chang-Tien Lu. Augmentation of chinese character representations with compositional graph learning (student abstract). In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pages 13075–13076, 2022.

[56] Lei Zhang, Zhiqian Chen, Chang-Tien Lu, and Liang Zhao. From "dynamics on graphs" to "dynamics of graphs": An adaptive echo-state network solution (student abstract). In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pages 13111–13112, 2022.

[57] Zonghan Zhang, Subhodip Biswas, Fanglan Chen, Kaiqun Fu, Taoran Ji, Chang-Tien Lu, Naren Ramakrishnan, and Zhiqian Chen. Blocking influence at collective level with hard constraints (student abstract). In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pages 13115–13116, 2022.

[58] Zhiqian Chen, Lei Zhang, Gaurav Kolhe, Hadi Mardani Kamali, Setareh Rafatirad, Sai Manoj Pudukotai Dinakarrao, Houman Homayoun, Chang-Tien Lu, and Liang Zhao. Deep graph learning for circuit deobfuscation. *Frontiers in big Data*, 4:608286, 2021.

[59] Kaiqun Fu, Taoran Ji, Nathan Self, Zhiqian Chen, and Chang-Tien Lu. A hierarchical attention graph convolutional network for traffic incident impact forecasting. In *2021 IEEE International Conference on Big Data (Big Data)*, pages 1619–1624. IEEE, 2021.

[60] Taoran Ji, Nathan Self, Kaiqun Fu, Zhiqian Chen, Naren Ramakrishnan, and Chang-Tien Lu. Dynamic multi-context attention networks for citation forecasting of scientific publications. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 7953–7960, 2021.

[61] Guoming Li, Yanbo Huang, Zhiqian Chen, Gary D Chesser Jr, Joseph L Purswell, John Linhoss, and Yang Zhao. Practices and applications of convolutional neural network-based computer vision systems in animal farming: A review. *Sensors*, 21(4):1492, 2021.

[62] Guoming Li, Xue Hui, Zhiqian Chen, Gary D Chesser Jr, and Yang Zhao. Development and evaluation of a method to detect broilers continuously walking around feeder as an indication of restricted feeding behaviors. *Computers and electronics in agriculture*, 181:105982, 2021.

[63] Beiyu Lin, Xiaowei Jia, and Zhiqian Chen. Sdm 21 tutorial: Studying spread patterns of covid-19 based on spatiotemporal data. In *SIAM Data Mining 2021*, 2021.

[64] Padmaksha Roy, Shailik Sarkar, Subhodip Biswas, Fanglan Chen, Zhiqian Chen, Naren Ramakrishnan, and Chang-Tien Lu. Deep diffusion-based forecasting of covid-19 by incorporating network-level mobility information. In *Proceedings of the 2021 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining*, pages 168–175, 2021.

[65] Subhodip Biswas, Fanglan Chen, Zhiqian Chen, Chang-Tien Lu, and Naren Ramakrishnan. Incorporating domain knowledge into memetic algorithms for solving spatial optimization problems. In *Proceedings of the 28th International Conference on Advances in Geographic Information Systems*, pages 25–35, 2020.

[66] Subhodip Biswas, Fanglan Chen, Andreea Sistrunk, Sathappan Muthiah, Zhiqian Chen, Nathan Self, Chang-Tien Lu, and Naren Ramakrishnan. Geospatial clustering for balanced and proximal schools. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 13358–13365, 2020.

[67] Fanglan Chen, Zhiqian Chen, Subhodip Biswas, Shuo Lei, Naren Ramakrishnan, and Chang-Tien Lu. Graph convolutional networks with kalman filtering for traffic prediction. In *Proceedings of the 28th international conference on advances in geographic information systems*, pages 135–138, 2020.

[68] Zhiqian Chen. Graph neural networks: Techniques and applications. 2020.

[69] Zhiqian Chen, Gaurav Kolhe, Setareh Rafatirad, Chang-Tien Lu, Sai Manoj PD, Houman Homayoun, and Liang Zhao. Estimating the circuit de-obfuscation runtime based on graph deep learning. In *2020 Design, Automation & Test in Europe Conference & Exhibition (DATE)*, pages 358–363. IEEE, 2020.

[70] Zhao Ding, Zhiqian Chen, Tianyi Ma, Chang-Tien Lu, Wenhui Ma, and Leon Shaw. Predicting the hydrogen release ability of libh4-based mixtures by ensemble machine learning. *Energy Storage Materials*, 27:466–477, 2020.

[71] Zhao Ding, Shaoyuan Li, Yang Zhou, Zhiqian Chen, Weijie Yang, Wenhui Ma, and Leon Shaw. Libh4 for hydrogen storage-new perspectives. *Nano Materials Science*, 2(2):109–119, 2020.

[72] Jianfeng He, Xuchao Zhang, Shuo Lei, Zhiqian Chen, Fanglan Chen, Abdulaziz Alhamadani, Bei Xiao, and ChangTien Lu. Towards more accurate uncertainty estimation in text classification. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 8362–8372, 2020.

[73] Subhodip Biswas, Fanglan Chen, Zhiqian Chen, Andreea Sistrunk, Nathan Self, Chang-Tien Lu, and Naren Ramakrishnan. Regal: A regionalization framework for school boundaries. In *Proceedings of the 27th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems*, pages 544–547, 2019.

[74] Taoran Ji, Zhiqian Chen, Nathan Self, Kaiqun Fu, Chang-Tien Lu, and Naren Ramakrishnan. Patent citation dynamics modeling via multi-attention recurrent networks. In *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence (IJCAI-19)*, 2019.

[75] Ying Zhang, Xingfeng He, Zhiqian Chen, Qiang Bai, Adelaide M Nolan, Charles A Roberts, Debasish Banerjee, Tomoya Matsunaga, Yifei Mo, and Chen Ling. Unsupervised discovery of solid-state lithium ion conductors. *Nature communications*, 10(1):5260, 2019.

[76] Zhiqian Chen, Feng Chen, Rongjie Lai, Xuchao Zhang, and Chang-Tien Lu. Rational neural networks for approximating graph convolution operator on jump discontinuities. In *2018 IEEE International Conference on Data Mining (ICDM)*, pages 59–68. IEEE, 2018.

[77] Kaiqun Fu, Zhiqian Chen, and Chang-Tien Lu. Streetnet: preference learning with convolutional neural network on urban crime perception. In *Proceedings of the 26th ACM SIGSPATIAL international conference on advances in geographic information systems*, pages 269–278, 2018.

[78] Manu Shukla, Zhiqian Chen, and Chang-Tien Lu. Dimpl: a distributed in-memory drone flight path builder system. *Journal of Big Data*, 5:1–29, 2018.

[79] Bingsheng Wang, Zhiqian Chen, Arnold P Boedihardjo, and Chang-Tien Lu. Virtual metering: An efficient water disaggregation algorithm via nonintrusive load monitoring. *ACM Transactions on Intelligent Systems and Technology (TIST)*, 9(4):1–30, 2018.

[80] Xuchao Zhang, Liang Zhao, Zhiqian Chen, and Chang-Tien Lu. Distributed self-paced learning in alternating direction method of multipliers. *arXiv preprint arXiv:1807.02234*, 2018.

[81] Zhiqian Chen, Chih-Wei Wu, Yen-Cheng Lu, Alexander Lerch, and Chang-Tien Lu. Learning to fuse music genres with generative adversarial dual learning. In *2017 IEEE International Conference on Data Mining (ICDM)*, pages 817–822. IEEE, 2017.

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